AI Model Types Explained

A simple, tabular guide to common AI model types, what they do, and where they are used.

Note: “Model type” can describe a model’s task, how it represents information, or its architecture. One AI application can combine several types.

Models that represent or generate content

Model type	What it does	Example use	Quick way to remember
Language models (LLMs)	Understand and generate text by predicting likely next words or tokens.	Chatbots, summarizing, writing, and question answering.	Generate and work with language.
Embedding models	Convert text, images, or other data into number lists (vectors) that capture meaning. Similar items have nearby vectors.	Semantic search, recommendations, and finding related documents.	Represent meaning as numbers.
Diffusion models	Generate content by starting with random noise and gradually refining it into a result.	Create or edit images from text prompts.	Build an image by gradually clearing noise.
Transformer models	Use attention to identify relationships between parts of the input. This is an architecture used in many language and multimodal systems.	Chatbots, translation, and document analysis.	A common design for processing connected information.
Encoder models	Read input and produce a useful representation, usually for understanding or analysis.	Text classification, sentiment analysis, and information extraction.	Read and represent input.
Decoder models	Generate output, commonly one token at a time.	Text generation and chatbot responses.	Produce output.
Encoder-decoder models	Read an input and generate a transformed output.	Translation and summarization.	Read input, then produce transformed output.
Classification models	Assign an input to one or more categories.	Spam detection or positive/neutral/negative sentiment.	Predict a label.
Regression models	Predict a numeric value.	Sales, demand, or house-price estimates.	Predict a number.
Clustering models	Group similar items without needing labels in advance.	Customer segments or groups of related documents.	Discover natural groups.
Recommendation models	Suggest items based on user preferences or similarities.	Recommend products, movies, or courses.	Suggest what may be useful or interesting.

Model type	What it does	Example use	Quick way to remember
Time-series models	Learn patterns in data collected over time to forecast future values.	Forecast energy use or product demand.	Use past values to estimate future values.
Graph models	Learn from entities and the relationships between them.	Fraud networks and social networks.	Understand connected things.
Multimodal models	Work with multiple kinds of information, such as text, images, audio, or video.	Ask questions about an image or describe a video.	Work across more than one data type.
Speech and audio models	Recognize, generate, or analyze speech and other audio.	Speech-to-text, text-to-speech, or sound classification.	Work with spoken words and sounds.
Vision models	Analyze images or video, such as detecting objects or reading text.	Identify objects in a photo or read a scanned page.	Interpret visual information.
Reinforcement-learning models	Learn actions through rewards and penalties from an environment.	Robotics, games, and sequential decision-making.	Learn by trying actions and receiving feedback.

How they can work together

A document-question chatbot can use an embedding model to find relevant passages and a language model to write an answer from those passages. That language model may use the transformer architecture. If a user adds a photo, a vision or multimodal model can interpret it.